

B.Tech IVth Semester Midterm Examination March 2025

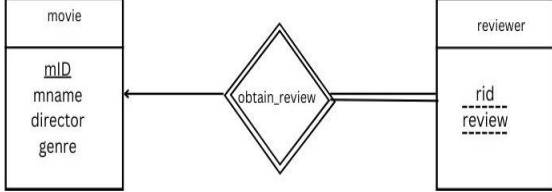
DATABASE SYSTEMS 2224

04/03/2025 8.30 AM TO 10.00 AM

30M

Q.No .	Question	Mark s	CLO/C O	AHE PLO	BT L
1	What is data redundancy in a database system? A. The process of normalizing data to remove duplicate records B. The unnecessary duplication of data within a database C. The intentional replication of data for backup purposes D. The practice of encrypting data for security reasons	0.5	1	1	2
2	Which of the following statements best describes the difference between Procedural DML and Declarative DML? A. Procedural DML specifies what data is needed, while Declarative DML specifies how to retrieve it. B. Procedural DML tells the database how to get the data step by step, while Declarative DML only tells what data is needed. C. Procedural DML is used only in relational databases, while Declarative DML is used in non-relational databases. D. There is no difference between Procedural DML and Declarative DML; both work the same way.	0.5	1	1	2
3	Which type of database user is responsible for defining the structure of the database and managing its overall design? A. End Users B. Database Administrator (DBA) C. Application Programmers D. Naïve Users	0.5	1	1	2
4	What is the primary purpose of a data dictionary in a database system? A. To store actual data records in tables B. To maintain metadata, such as table structures and constraints C. To back up the entire database automatically D. To encrypt sensitive data for security purposes	0.5	1	1	2

5	--- is not a basic operator in relational algebra. A. set difference: – B. rename: ρ C. Cartesian product: x D. Set intersection : ∩	0.5	1	1	2										
6	Consider the entity relationship which consists of two entity sets, customer and loan, related through a binary relationship set borrower. An undirected line from the relationship set borrower to the entity set loan specifies that borrower is either a _____ or _____ relationship set from customer to loan. . One to one, many to one . many to many, one to many . Many to many, one to one D. One to One, Many to Many	0.5	3	5	3										
7	Match the following <table><tr><th>Symbol</th><th>Name</th></tr><tr><td>i. σ</td><td>v. Cartesian Product</td></tr><tr><td>ii. π</td><td>vi. Natural Join</td></tr><tr><td>iii. ⋈</td><td>vii. Projection</td></tr><tr><td>iv. ×</td><td>viii. Selection</td></tr></table> a. i – viii, ii – vii, iii – vi, iv - v b. i – vii, ii – viii, iii – vi, iv - v c. i – v, ii – vii, iii – vi, iv - viii d. i – viii, ii – vi, iii –vii, iv - v	Symbol	Name	i. σ	v. Cartesian Product	ii. π	vi. Natural Join	iii. ⋈	vii. Projection	iv. ×	viii. Selection	0.5	1	1	1
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8	A B What type of relationship exists between the A and B entities in the given E-R diagram? A) One-to-One (1:1) B) One-to-Many (1:M) C) Many-to-One (M:1)	0.5	3	5	2										

	D) Many-to-Many (M:N)				
9	In Fig 1, The reviewer entity set is -----  a) strong b) weak c) semi strong d) super	0.5	3	5	2
10	----- is an abstraction through which relationships are treated as higher-level entities. a) Aggregation b) Specialization c) Generalization d) Inheritance	0.5	3	5	2

11	Suppose a relational schema $R(a,b,c,z)$ and set of functional dependency as following: $F: \{b \rightarrow a, az \rightarrow bc, c \rightarrow abz\}$. Find the canonical cover F_c. Ans: For this we will decompose all the FD $b \rightarrow a, az \rightarrow b, az \rightarrow c, c \rightarrow a, c \rightarrow z, c \rightarrow b$ For RHS: Considering $b^+ = b$ without using $b \rightarrow a$ so $b \rightarrow a$ is essential, 0.5M $az^+ = azcb$ without using $az \rightarrow b$ so it is not essential 0.5M $az^+ = az$ without using $az \rightarrow c$ so it is essential 0.5M $c^+ = cbza$ without $c \rightarrow a$ so $c \rightarrow a$ is not essential 0.5M $c^+ = cz$ without $c \rightarrow b$ so $c \rightarrow b$ is essential 0.5M $c^+ = cxa$ without $c \rightarrow z$ so $c \rightarrow z$ is essential 0.5M So $F: \{ b \rightarrow a, az \rightarrow c, c \rightarrow z, c \rightarrow b \}$ For LHS: Find wz^+ using F and if we check any one attribute is extra $a^+ \rightarrow a$ $z^+ \rightarrow z$ so since y is not derived by anyone of the attribute so $az \rightarrow c$ is essential 0.5M therefore minimal cover is	4	4	2	4
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	$F_c: \{b \rightarrow a, az \rightarrow c, c \rightarrow z, c \rightarrow b\}$ 0.5M (Only if the result is right 0.5M will be awarded) So if a student has any one FD wrong then definitely last answer will be wrong , so he/she will get max 3M				
12	Discuss the functionalities of different components of query processor and storage manager Ans: Storage Manager ❖ Storage manager is a component of database/program module that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system. ❖ The storage manager is responsible to the following tasks: ◦ Interaction with the OS file manager ◦ Efficient storing, retrieving and updating of data ❖ Issues: ◦ Storage access ◦ File organization ◦ Indexing and hashing ❖ The storage manager components include: ◦ • Authorization and integrity manager ◦ • Transaction manager ◦ • File manager ◦ • Buffer manager Implements several data structures as part of the physical system implementation: Data Files, Data Dictionary and Indices Query Processor ❖ Parsing and translation, Optimization ,Evaluation. ❖ The query processor components include: • DDL interpreter, interprets DDL statements, records definitions in the data dictionary. • DML compiler, translates DML statements in a query language into an evaluation plan consisting of low-level instructions that the query evaluation engine understands. • The DML compiler also performs query optimization; that is, it picks the lowest cost evaluation plan from among the alternatives. • Query evaluation engine, which executes low-level instructions generated by the DML compiler. ❖ Alternative ways of evaluating a given query • Equivalent expressions • Different algorithms for each operation ❖ Cost difference between a good and a bad way of evaluating a query can be enormous ❖ Need to estimate the cost of operations • Depends critically on statistical information about relations which the database must maintain • Need to estimate statistics for intermediate results to compute cost of complex expressions	3	1	1	2

13	Consider the following schema: Employee(Fname,Mname,Lname,<u>Essn</u>,bdate,address,salary,Dnu mber) Department(Dname, <u>Dnumber</u>, Mgr_ssn,Mgr_start_date) Dependent(<u>Essn</u>,<u>Dependent_name</u>,gender,Relationship) write Relational algebra expressions for the following questions: a) Retrieve the Fname, Lname and address of all employees who work for the 'Research' department. b) Find the average salary of employees in each department. Retrieve the details of employees who have more than two dependents Ans: (1*1→3 Marks) 1. Retrieve the Fname, Lname and address of all employees who work for the 'Research' department. Ans: $\pi_{Fname,Lname,Address}(\sigma_{Dname='Research'}(Depa$ rtment $\bowtie$ Employee)) 2. Find the average salary of employees in each department. Ans: $\pi_{Dname, av(Dname \bowtie$ avg(salary)as av(Employee $\bowtie$ Department)) 3. Retrieve the details of employees who have more than two dependents Ans: Temp1->essn $\bowtie$ count(Dependent_name)>2(Employee $\bowtie$ Depende nt) Π Fname, Lname(Temp1 $\bowtie$ Employee)	3	1	1	3
14	Consider the E-R diagram for a banking enterprise provided below. (a) List the cardinality and participation for each relationship in the diagram. (b) Convert the E-R diagram into a relational schema.	3	3	5	4

	Scheme: Cardinality & participation – 1m 4 tables for Entities -1m 2 tables for Relationships -1m <table><tr><th>Relationship</th><th>Cardinality</th><th>Participation</th></tr><tr><td>Account – Branch</td><td>Many - One</td><td>Total (Account) – Partial (Branch)</td></tr><tr><td>Loan – Branch</td><td>Many - One</td><td>Total (Total) – Partial (Branch)</td></tr><tr><td>Depositor</td><td>Many – Many</td><td>Partial – Partial</td></tr><tr><td>Borrower</td><td>Many – Many</td><td>Partial – Partial</td></tr></table> ■ The following relational schemes result: <pre>branch (branch-name, branch-city, assets) customer (customer-name, customer-street, customer-city) account (account-number, branch-name, balance) loan (loan-number, branch-name, amount) depositor (customer-name, account-number) borrower (customer-name, loan-number)</pre>	Relationship	Cardinality	Participation	Account – Branch	Many - One	Total (Account) – Partial (Branch)	Loan – Branch	Many - One	Total (Total) – Partial (Branch)	Depositor	Many – Many	Partial – Partial	Borrower	Many – Many	Partial – Partial				
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Borrower	Many – Many	Partial – Partial																		
15	Raju wishes to model an inventory management system with customers, employees and products, that need to be managed. Every customer information stored comes with ID, cname, and a mandatory purchase of atleast one product with purchase date. A customer can buy a maximum of k products and a product can be bought by 1 or m customers. Every product has ID, title and description. Every employee is either a regular employee or a manager, who manages 1 to m products where as a product can be managed exactly one employee. Considering the given scenario, identify primary key, cardinality as low..high with necessary roles of employee, illustrate the scenario with a neat ER model. Solution: Cardinality 1M ER diagram with primary key 1M Roles and total participation 1M	3	3	5	3															

16	Consider the following schema for Airline Reservation System □ Passengers (<u>PassengerID</u>, Name, ContactInfo) □ Reservations (<u>ReservationID</u>, PassengerID, FlightID, SeatNo, BookingDate) □ Flights (<u>FlightID</u>, Airline, Source, Destination, DepartureTime, ArrivalTime) □ Airlines (<u>AirlineID</u>, Name, Country) Using 'with clause' formulate a query to find the FlightId, Name and number of passengers who have made a reservation but have not been assigned a seat for their flight. Display the flight ID, airline name, and the total count of unassigned reservations per flight. Ans: <pre>WITH UnassignedReservations AS (SELECT r.FlightID, f.Airline, a.Name AS AirlineName FROM Reservations r JOIN Flights f ON r.FlightID = f.FlightID JOIN Airlines a ON f.Airline = a.AirlineID WHERE r.SeatNo IS NULL) SELECT FlightID, AirlineName, COUNT(*) AS UnassignedPassengers FROM UnassignedReservations GROUP BY FlightID, AirlineName</pre>	3	2	2	3
17	Using the airline schema in Q16, write SQL query to Retrieve the names of passengers, the airline name, the source and destination of their flight, and their seat number for flights traveling from New York to Los Angeles. Display the results sorted by airline name. Ans: <pre>SELECT p.Name AS PassengerName, a.Name AS AirlineName, f.Source, f.Destination, r.SeatNo FROM Reservations r JOIN Passengers p ON r.PassengerID = p.PassengerID JOIN Flights f ON r.FlightID = f.FlightID JOIN Airlines a ON f.Airline = a.AirlineID WHERE f.Source = 'New York' AND f.Destination = 'Los Angeles' ORDER BY a.Name; 0.5M</pre> Scheme: Answer written without join condition 1M	2	2	2	3
18	Consider the following schema and write SQL queries for the following: Library (<u>Libname</u>, city) book(<u>ISBN</u>, Libname, title, price) a) Using an appropriate join operator, list all the library names with the number of books they have. b) Rewrite the following query using join operator: (Select Libname from library) except (select distinct(Libname) from book)	2	2	2	

	Ans: a) select Libname, count(ISBN) from library natural left outer join book group by Libname; b) select Libname from library natural left outer join book where ISBN is null;				
19	Describe the ways to create user-defined types in SQL. Ans: i) create type clause can be used to define new types. For example, the statements: create type Dollars as numeric(12,2) final; define the user-defined types Dollars to be decimal number. The newly created types can then be used as types of attributes of relations. For example, we can declare the department table as: create table department (dept name varchar (20), building varchar (15), budget Dollars);-----1M ii) we could define a domain DDollars as follows.: create domain DDollars as numeric(12,2) not null; The domain DDollars can be used as an attribute type, just as we used the type Dollars.-----1M	2	2	2	